Syllabus

Introduction

Let $X_1, \ldots, X_n$ be iid P_θ.
Let $\hat\theta = \hat\theta(X_1, \ldots, X_n)$ be an estimator for θ.

One often wants to evaluate $\mathrm{Var}[\hat\theta]$ (to quantify the uncertainty of $\hat\theta$).

The bootstrap is a powerful, broadly applicable, method

- to estimate $\mathrm{Var}[\hat\theta]$,
- to estimate the bias $\mathrm{E}[\hat\theta] - \theta$ of $\hat\theta$,
- to construct confidence intervals for θ,
- to...

The method is nonparametric and can deal with small n.

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An example from James et al. (2013)

Let Y, Z be the values of two random assets and consider the portfolio

$$W_\lambda = \lambda Y + (1-\lambda) Z, \qquad \lambda \in [0,1],$$

allocating a proportion λ of your wealth to Y and a proportion $1-\lambda$ to Z. A common, risk-averse, strategy is to minimize the risk $\mathrm{Var}[W_\lambda]$.

It can be shown that this risk is minimized at

$$\lambda_{\mathrm{opt}} = \frac{\mathrm{Var}[Y] - \mathrm{Cov}[Y,Z]}{\mathrm{Var}[Y] + \mathrm{Var}[Z] - 2\mathrm{Cov}[Y,Z]}.$$

But in practice, $\mathrm{Var}[Y]$, $\mathrm{Var}[Z]$ and $\mathrm{Cov}[Y,Z]$ are unknown...

Sample case

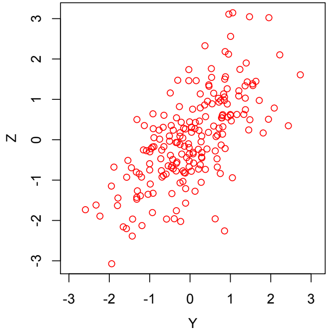
Now, if historical data $X_1 = (Y_1, Z_1), \ldots, X_n = (Y_n, Z_n)$ are available, then we can estimate λ_{opt} by

$$\hat{\lambda}_{\mathrm{opt}} = \frac{\widehat{\mathrm{Var}[Y]} - \widehat{\mathrm{Cov}[Y,Z]}}{\widehat{\mathrm{Var}[Y]} + \widehat{\mathrm{Var}[Z]} - 2\widehat{\mathrm{Cov}[Y,Z]}},$$

where

- $\widehat{\mathrm{Var}[Y]}$ is the sample variance of the Y_i's
- $\widehat{\mathrm{Var}[Z]}$ is the sample variance of the Z_i's
- $\widehat{\mathrm{Cov}[Y,Z]}$ is the sample covariance of the Y_i's and Z_i's.

Sample case



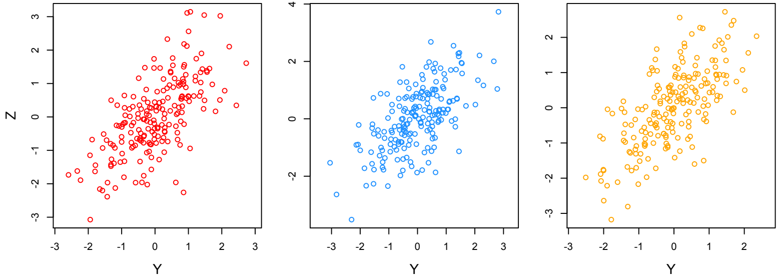
For this sample, $\hat{\lambda}_{\rm opt} = .283$.

How to estimate the accuracy of $\hat{\lambda}_{\rm opt}$ (i.e., its standard deviation $\mathrm{Std}[\hat{\lambda}_{\rm opt}]$)?

On the basis of the available sample, we observe $\hat{\lambda}_{\rm opt}$ only once.
$\rightsquigarrow$ We need further samples leading to further observations of $\hat{\lambda}_{\rm opt}$.

Sampling from the population

We generated 1000 samples from the population. The first three are



$\rightsquigarrow$ $\hat{\lambda}^{(1)}_{\text{opt}} = .283$, $\hat{\lambda}^{(2)}_{\text{opt}} = .357$, $\hat{\lambda}^{(3)}_{\text{opt}} = .299$, ..., which allows us to compute

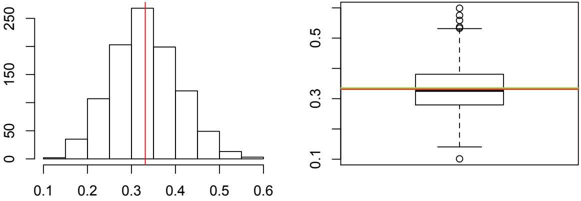
$$\widehat{\text{Std}[\hat{\lambda}_{\text{opt}}]} = \sqrt{\frac{1}{999}\sum_{i=1}^{1000}(\hat{\lambda}^{(i)}_{\text{opt}} - \bar{\lambda}_{\text{opt}})^2}, \quad \text{with } \bar{\lambda}_{\text{opt}} = \frac{1}{1000}\sum_{i=1}^{1000}\hat{\lambda}^{(i)}_{\text{opt}}.$$

Sampling from the population

Here,

$$\widehat{\mathrm{Std}[\hat{\lambda}_{\mathrm{opt}}]} \approx .077, \qquad \bar{\lambda}_{\mathrm{opt}} \approx .331 \;\; (\approx \lambda_{\mathrm{opt}} = 1/3 = .333),$$

and the distribution of $\hat{\lambda}_{\mathrm{opt}}$ is described by



(This could also be used to estimate quantiles of $\hat{\lambda}_{\mathrm{opt}}$.)

Sampling from the sample: the bootstrap

It is important to realize that this cannot be done in practice (one cannot sample from the population P_θ since it is unknown).

However, one may sample instead from the empirical distribution P_n (i.e., the uniform distribution over $X_1, \ldots, X_n$), that is close to P_θ for large n!

This means that we sample with replacement from $X_1, \ldots, X_n$, providing a first bootstrap sample $(X_1^{*1}, \ldots, X_n^{*1})$, which allows us to evaluate $\hat{\lambda}_{\text{opt}}^{*(1)}$. Further generating bootstrap samples $(X_1^{*b}, \ldots, X_n^{*b})$, $b = 2, \ldots, B = 1000$, one can compute

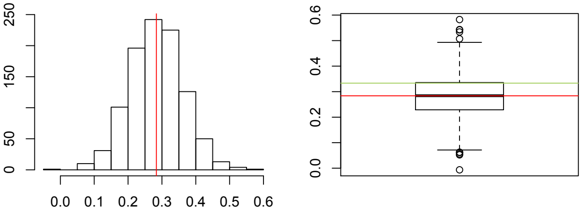
$$\widehat{\mathrm{Std}[\hat{\lambda}_{\text{opt}}]}^{\,*} = \sqrt{\frac{1}{B-1} \sum_{b=1}^{B} (\hat{\lambda}_{\text{opt}}^{*(b)} - \bar{\lambda}_{\text{opt}}^{*})^2}, \quad \text{with } \bar{\lambda}_{\text{opt}}^{*} = \frac{1}{1000} \sum_{b=1}^{B} \hat{\lambda}_{\text{opt}}^{*(b)}.$$

Sampling from the sample: the bootstrap

This provides

$$\widehat{\mathrm{Std}[\hat{\lambda}_{\mathrm{opt}}]}^{\,*} \approx .079,$$

and the distribution of $\hat{\lambda}_{\mathrm{opt}}$ is described by

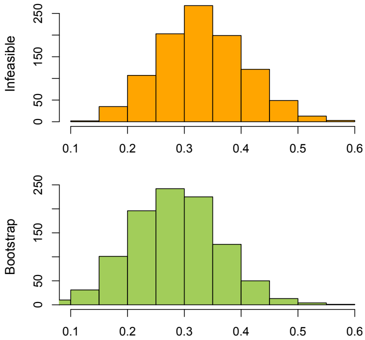


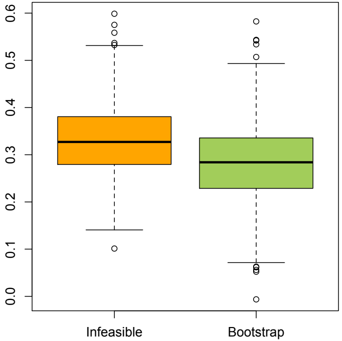
(This could again be used to estimate quantiles of $\hat{\lambda}_{\mathrm{opt}}$.)

A comparison between both samplings

Results are close!

$$\widehat{\mathrm{Std}}[\hat{\lambda}_{\mathrm{opt}}] \approx .077 \quad \text{and} \quad \widehat{\mathrm{Std}}[\hat{\lambda}_{\mathrm{opt}}]^* \approx .079$$





Syllabus

1. The bootstrap
   - A motivating example
   - The general procedure
   - About the implementation in R
2. Bagging of trees
3. Random forests (an improved bagging of trees)

The bootstrap

Let $X_1, \ldots, X_n$ be iid P_θ.
Let $T = T(X_1, \ldots, X_n)$ be a statistic of interest.
The bootstrap allows us to say something about the distribution of T:

$$
\begin{array}{ccc}
(X_1^{*1}, \ldots, X_n^{*1}) & \rightsquigarrow & T^{*1} = T(X_1^{*1}, \ldots, X_n^{*1}) \\
 & \vdots & \\
(X_1^{*b}, \ldots, X_n^{*b}) & \rightsquigarrow & T^{*b} = T(X_1^{*b}, \ldots, X_n^{*b}) \\
 & \vdots & \\
(X_1^{*B}, \ldots, X_n^{*B}) & \rightsquigarrow & T^{*B} = T(X_1^{*B}, \ldots, X_n^{*B})
\end{array}
$$

Under mild conditions, the empirical distribution of $T^{*1}, \ldots, T^{*B}$ provides a good approximation of the sampling distribution of T under P_θ.

The bootstrap

Above, each bootstrap sample $(X_1^{*b}, \ldots, X_n^{*b})$ is obtained by sampling (uniformly) with replacement among the original sample $(X_1, \ldots, X_n)$.

Possible uses:

- $\frac{1}{n-1} \sum_{b=1}^{B} (T^{*b} - \bar{T}^*)^2$, with $\bar{T}^* = \frac{1}{n} \sum_{b=1}^{B} T^{*b}$, estimates $\mathrm{Var}[T]$
- The sample α-quantile q_α^* of $T^{*1}, \ldots, T^{*B}$ estimates T's α-quantile

Possible uses when T is an estimator of θ:

- $(\frac{1}{n} \sum_{b=1}^{B} T^{*b}) - T$ estimates the bias $\mathrm{E}[T] - \theta$ of T
- $[q^*_{\alpha/2}, q^*_{1-(\alpha/2)}]$ is an approximate $(1 - \alpha)$-confidence interval for θ.
- ...

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A toy illustration

Let $X_1, \ldots, X_n$ ($n = 4$) be iid t-distributed with 6 degrees of freedom.
Let $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean.
How to estimate the variance of $\bar{X}$ through the bootstrap?

```
n <- 4
X <- rt(n,df=6)
X

## [1] -0.08058779  0.28044078  1.19011050 -1.25212790
```

```
Xbar <- mean(X)
Xbar

## [1] 0.0344589
```

Obtaining a bootstrap sample

```
X
```

```
## [1] -0.08058779  0.28044078  1.19011050 -1.25212790
```

```
d <- sample(1:n,n,replace=TRUE)
d
```

```
## [1] 2 4 4 4
```

```
Xstar <- X[d]
Xstar
```

```
## [1]  0.2804408 -1.2521279 -1.2521279 -1.2521279
```

Generating $B = 1000$ bootstrap means

```r
B <- 1000
Bootmeans <- vector(length=B)
for (b in (1:B))
     {
     d <- sample(1:n,n,replace=TRUE)
     Bootmeans[b] <- mean(X[d])
     }
Bootmeans[1:4]
```

```
## [1]  0.2370868 -0.3486833  0.3521335  0.2370868
```

Bootstrap estimates

Bootstrap estimates of $\mathrm{E}[\bar{X}]$ and $\mathrm{Var}[\bar{X}]$ are then given by

```
mean(Bootmeans)

## [1] 0.03679914

var(Bootmeans)

## [1] 0.1789107
```

The practical sessions will explore how well such estimates behave.

The boot function

A better strategy is to use the `boot` function from

```r
library(boot)
```

The `boot` function takes typically 3 arguments:

- `data`: the original sample
- `statistic`: a user-defined function with the statistic to bootstrap
- `R`: the number B of bootstrap samples to consider

The user-defined function describes how the statistic is computed from

- (1st argument:) a generic sample, and
- (2nd argument:) a vector of indices pointing to a subsample on which the statistic is to be evaluated...

The boot function

If the statistic is the mean, then a suitable user-defined function is

```
boot.mean <- function(x,d) {
  mean(x[d])
}
```

The bootstrap estimate of $\mathrm{Var}[\bar{X}]$ is then

```
res.boot <- boot(X,boot.mean,R=1000)
var(res.boot$t)

##           [,1]
## [1,] 0.1844024
```